Awesome 🚀 Let's continue with **Module 13: GROUP BY**.

📘 SQL CLASS NOTE – MODULE 13: `GROUP BY`

1. Introduction

The **`GROUP BY` clause** is used to **arrange rows into groups** based on one or more columns.
It is usually combined with **aggregate functions** (`COUNT` , `SUM` , `AVG` , `MIN` , `MAX`).

👉 Without `GROUP BY` , aggregate functions summarize the **entire table**.
👉 With `GROUP BY` , they summarize **each group separately**.

2. Syntax

```
SELECT column1, AGG_FUNCTION(column2)
FROM table_name
WHERE condition
GROUP BY column1;
```

3. Practical Examples with `SchoolDB`

Example 1: Count students per grade

```
SELECT GradeLevel, COUNT(*) AS StudentCount
FROM Students
GROUP BY GradeLevel;
```

Example 2: Find average age of students per grade

```sql
SELECT GradeLevel, AVG(Age) AS AvgAge
FROM Students
GROUP BY GradeLevel;
```

Example 3: Find number of students by gender

```sql
SELECT Gender, COUNT(*) AS Total
FROM Students
GROUP BY Gender;
```

Example 4: Find the youngest and oldest student per grade

```sql
SELECT GradeLevel, MIN(Age) AS Youngest, MAX(Age) AS Oldest
FROM Students
GROUP BY GradeLevel;
```

Example 5: Count enrollments per course

```sql
SELECT CourseID, COUNT(*) AS EnrollmentCount
FROM Enrollments
GROUP BY CourseID;
```

Example 6: Find number of courses taught by each teacher

```sql
SELECT Teacher, COUNT(*) AS CourseCount
FROM Courses
GROUP BY Teacher;
```

Example 7: Find average student age per gender

```
SELECT Gender, AVG(Age) AS AvgAge
FROM Students
GROUP BY Gender;
```

4. Why Use `GROUP BY` ?

- Helps summarize **categories** of data.
- Works hand-in-hand with aggregate functions.
- Makes reports **more meaningful**.

5. Real-Life Use Cases

- **Schools**: Count students per grade.
- **Banks**: Sum transactions per account type.
- **Hospitals**: Number of patients per department.
- **E-commerce**: Sales total per category.

6. Classwork (Practice) ✅

👉 Try these queries:

1. Show the number of students in each **GradeLevel**.
2. Find the average age of students per **GradeLevel**.
3. Count the number of students grouped by **Gender**.
4. Show the minimum and maximum student ages per **GradeLevel**.
5. Count the number of enrollments per **CourseID**.
6. Show the number of courses taught by each **Teacher**.
7. Find the average student age grouped by **Gender**.

7. Challenge Question 💡

👉 Write a query to display:

- Each `GradeLevel`
- Number of male students
- Number of female students

📌 (Hint: Use `GROUP BY` with `CASE WHEN` for gender count).

✅ That's the **expanded Module 13: GROUP BY**.

Perfect 👍 Let's solve the **Module 13 Classwork + Challenge** step by step with SQL queries.

✅ Solutions to Module 13: `GROUP BY`

1. Show the number of students in each **GradeLevel**.

```
SELECT GradeLevel, COUNT(*) AS StudentCount
FROM Students
GROUP BY GradeLevel;
```

2. Find the average age of students per **GradeLevel**.

```
SELECT GradeLevel, AVG(Age) AS AvgAge
FROM Students
GROUP BY GradeLevel;
```

3. Count the number of students grouped by **Gender**.

```
SELECT Gender, COUNT(*) AS TotalStudents
FROM Students
GROUP BY Gender;
```

4. Show the minimum and maximum student ages per **GradeLevel**.

```
SELECT GradeLevel,
       MIN(Age) AS YoungestAge,
       MAX(Age) AS OldestAge
FROM Students
GROUP BY GradeLevel;
```

5. Count the number of enrollments per **CourseID**.

```
SELECT CourseID, COUNT(*) AS EnrollmentCount
FROM Enrollments
GROUP BY CourseID;
```

6. Show the number of courses taught by each **Teacher**.

```
SELECT Teacher, COUNT(*) AS CourseCount
FROM Courses
GROUP BY Teacher;
```

7. Find the average student age grouped by **Gender**.

```
SELECT Gender, AVG(Age) AS AvgAge
FROM Students
GROUP BY Gender;
```

💡 Challenge Question

👉 Display each `GradeLevel` with:

- Number of **male students**
- Number of **female students**

```
SELECT GradeLevel,
       SUM(CASE WHEN Gender = 'Male' THEN 1 ELSE 0 END) AS MaleStudents,
       SUM(CASE WHEN Gender = 'Female' THEN 1 ELSE 0 END) AS FemaleStudents
FROM Students
GROUP BY GradeLevel
ORDER BY GradeLevel ASC;
```

📌 Here we used **`CASE WHEN` inside `SUM()`** to count separately for Male and Female students.